

# Victor Amarante Florentino da Silva

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## PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience building production-grade intelligent systems — from classical ML and statistical modeling to LLM orchestration, agentic architectures, and RAG pipelines deployed at scale across fintech, healthcare, legal, and logistics. Grounded in Statistics and Probability Theory, with the mathematical depth to understand what happens inside a model and the engineering discipline to build systems that ship. Currently designing autonomous AI agents at one of Brazil’s largest financial institutions — owning the full model lifecycle from conception to production monitoring.

## EDUCATION

- **Bachelor’s Degree in Information Systems**

Federal University of Pernambuco (UFPE) - Center of Informatics (CIn)

Mar 2023 – Dec 2026  
Recife, Brazil

• **Bachelor’s Degree in Statistics (internal transfer)**

Federal University of Pernambuco (UFPE) - Center of Exact and Natural Sciences (CCEN)

Mar 2020 – Feb 2023  
Recife, Brazil

◦ **Internal Transfer:** Formal training in probability, applied statistics, linear algebra, and numerical calculus, the mathematical backbone behind machine learning and AI systems. Transferred internally to Information Systems to deepen competencies in Artificial Intelligence, Software Engineering, and System Design.
- ## EXPERIENCE
- **Senior Machine Learning Engineer**

Getronics

May 2026 – Present  
Amsterdam, Netherlands (Remote)

◦ Embedded at one of Brazil’s largest financial institutions as part of a high-seniority squad, owning the full AI model lifecycle — from architecture design to production monitoring — for a card services digital assistant projected to reach millions of users at launch.

◦ Architected a multi-agent system composed of an orchestrator and a specialist agent deployed as independent microservices, communicating via A2A protocol — designed for extensibility as additional agents and tools are integrated.

◦ Built and consumed REST APIs with MCP and A2A integrations, ensuring seamless connectivity across multiple enterprise systems.

◦ Enforced AI compliance controls across the full pipeline — input/output filtering, PII handling, and data privacy requirements — in adherence to Brazilian financial sector regulations.

◦ Drove prompt engineering practices including safety guardrails, temperature tuning, and token optimization to reduce latency and inference costs at scale.

◦ Ensured production observability and reliability through OpenTelemetry-based distributed tracing.

• **AI Researcher**

CIn/UFPE — Center of Informatics

Mar 2026 – Present  
Recife, Brazil

◦ Conducting applied research on "Role-Specialized Small Language Models for End-to-End Spec-Driven Software Development" — investigating whether SLMs ( $\leq 10B$  parameters) fine-tuned via LoRA per agent role can match frontier LLM performance in a full software development pipeline, with advantages in cost, latency, and on-premise execution.

◦ Architected a production-grade multi-agent system (Zup SDD Agents) in partnership with Zup Innovation: a LangGraph state machine with  $\approx 19$  nodes that transforms a User Story into functional, committed code — covering spec generation, supervised planning, specialist execution (backend, frontend, QA), technical verification, and business curation.

◦ Designed a dual quality gate separating technical verification (test evidence, diffs, logs) from business curation (User Story  $\leftrightarrow$  SPEC  $\leftrightarrow$  implementation alignment), with human-in-the-loop checkpoints via LangGraph’s interrupt mechanism.

◦ Implemented deep observability infrastructure — structured AgentReasoningEvents persisted per agent handoff, SSE real-time streaming, and optional LangSmith tracing — forming the trajectory dataset for future LoRA distillation and epistemic cascade degradation measurement.

• **Machine Learning Engineer**

Amigo Tech

Sep 2025 – May 2026  
Recife, Brazil

◦ Led AI development on Amigo Flow, architecting a supervised multi-agent system — orchestrator + specialist agents + dedicated RAG pipelines — applied to the patient journey, delivering a 7.5 $\times$  increase in automated scheduling and 200%+ growth in confirmation rate across tens of thousands of monthly appointments.

◦ Redesigned the core AI product architecture around a multi-agent pattern, achieving 90% improvement in production response quality and accuracy.

◦ Built an end-to-end medical imaging pipeline for dental X-ray analysis: annotated training data, trained YOLO-based object detection models, and integrated a VLM to generate structured diagnostic reports in natural language.

◦ Engineered prompt lifecycle and governance workflows with version-controlled repositories, applying iterative evaluation and token consumption analysis to improve quality and reduce operational costs.

◦ Implemented conversational monitoring and distributed tracing with LangSmith, enabling behavioral inspection and reliability improvements across production agent systems.

• **Data Scientist**

C.E.S.A.R – Center for Advanced Studies and Systems of Recife

Jul 2024 – Sep 2025  
Recife, Brazil

- Designed a GenAI-powered enterprise knowledge retrieval system combining semantic search with contextual response generation — built with LangChain/LangGraph, PostgreSQL, and pgvector.
  - Architected a GraphRAG system on Neo4j for a logistics data platform, combining vector search with entity relationship traversal to enable intelligent queries across previously siloed enterprise data.
  - Built a data integration platform for the transportation sector with an LLM-enriched ingestion pipeline on AWS S3 and assisted knowledge graph construction.
  - Deployed containerized AI pipelines on AWS (S3, SageMaker, Bedrock) and GCP (Vertex AI), focusing on scalable architecture and enterprise data infrastructure integration.
  - Delivered 20+ hours of in-person data analysis training for SEBRAE analysts (Natal/RN), driving data literacy and tooling adoption across the organization.
- **Data Scientist** Nov 2022 – Jun 2024  
*Queiroz Cavalcanti Advocacia* Recife, Brazil
- Built predictive ML pipelines for Legal Controlling — time series forecasting and classification models — delivering 90%+ efficiency gains across 50+ automation projects, saving an average of 3–4 hours per optimized process and up to 90 hours/month on a single initiative alone.
  - Designed analytical pipelines, statistical models, and Power BI dashboards for strategic KPI monitoring, significantly reducing operational overhead across multiple sub-teams of the legal controlling department.
  - Conceived and led an internal training program in automation and data science with 80%+ adoption across 40+ professionals.
  - Prototyped early generative AI solutions — automated meeting summarization and intelligent email prioritization — establishing the firm's AI foundation during the initial GenAI adoption wave.
- **Data Scientist** Jul 2021 – Jul 2023  
*Fisher Consultoria Estatística* Recife, Brazil
- Delivered 10+ end-to-end statistical consulting and data science projects across healthcare clients — from problem modeling and sampling design to analysis, interpretation, and formal report delivery.
  - Applied statistical modeling, hypothesis testing, and exploratory data analysis to solve quantitative business problems across diverse domains.
  - Communicated findings through data visualizations and analytical reports tailored to non-technical stakeholders.
- **Co-Founder & Director of Education** Dec 2024 – Mar 2026  
*LIGIA – Academic League of Artificial Intelligence at UFPE* Recife, Brazil
- Co-founded and led LIGIA, structuring learning tracks in Machine Learning, NLP, and Computer Vision for 30–40 active members, with additional reach through training programs delivered to external academic entities.
  - Developed educational content including introductory courses, teaching materials, and audiovisual production for AI training programs.
  - Mentored student leaders, fostering a culture of technical excellence, autonomy, and continuous learning.
- **Co-Founder & ML Engineer** Nov 2023 – Jul 2024  
*Redduo.AI* Recife, Brazil
- Co-founded and built an AI-powered educational platform for automatic, personalized essay correction for Brazil's ENEM exam, using LLMs and semantic analysis to generate structured student feedback at scale.
  - Designed the full data pipeline and backend architecture with cloud service integration and language model APIs, ensuring evaluation consistency and processing scalability.
  - Applied NLP, generative AI, and automated text analysis validated with linguistic and semantic metrics using supervised ML techniques.
  - **1st place at PoloTec Pre-Incubation (UFPE) and 1st place at PoloTec DemoDay 2024.**
- **Co-founder & AI Researcher** Jan 2024 – Mar 2026  
*GERAIA – Research and Applications Group in Generative AI* Recife, Brazil
- Conducted applied research on LLMs, multi-agent architectures, and RAG systems, with focus on autonomous simulations and distributed intelligent systems.
  - Developed AI agents with contextual memory, autonomous reasoning, and generative capabilities for dynamic real-world environments.
  - Optimized RAG pipelines and LLM integrations for semantic retrieval, information synthesis, and language-based process automation.
- **Scientific Researcher** Jul 2023 – Jan 2024  
*FACEPE – Foundation for Science and Technology Support of Pernambuco* Recife, Brazil
- Conducted research at INES (National Institute for Software Engineering) on accessible mobile applications for individuals with Autism Spectrum Disorder (ASD).
  - Applied Human-Computer Interaction (HCI), User-Centered Design, and data analysis methodologies for digital usability and accessibility evaluation.

SKILLS

- **AI Agents & LLM Systems** LangChain, LangGraph, LlamaIndex, Agno, CrewAI, RAG & GraphRAG, MCP, A2A, multi agent architectures, orchestrator-specialist patterns, ReAct, contextual memory, prompt engineering, fine-tuning
- **ML & Statistical Foundations** Scikit-learn, PyTorch, TensorFlow, MLflow, Pandas, NumPy, SciPy, Statsmodels, predictive modeling, time series, feature engineering, probability, applied statistics, linear algebra
- **Vector Search & Data** FAISS, ChromaDB, PGVector, Neo4j, PostgreSQL, SQL, SQLAlchemy, semantic & hybrid search
- **LLMOps & Observability** LangSmith, OpenTelemetry, distributed tracing, inference optimization, token cost analysis, evaluation pipelines, model monitoring
- **Cloud & Infrastructure** AWS (S3, SageMaker, Bedrock), GCP (Vertex AI), Docker, Kubernetes, FastAPI, Python, TypeScript

LICENSES & CERTIFICATIONS

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|---------------------------------------------------------------------|----------|
| • Building Agents with Agno (Asimov Academy)                        | Aug 2025 |
| • AI Applications: Multi-Agent Systems with CrewAI (Asimov Academy) | Jun 2025 |
| • Knowledge Graphs for RAG (DeepLearning.AI)                        | Mar 2025 |
| • Developing Chatbots (Rocketseat)                                  | Feb 2025 |
| • Functions, Tools and Agents with LangChain (DeepLearning.AI)      | Sep 2024 |
| • LangChain for LLM Application Development (DeepLearning.AI)       | Sep 2024 |
| • AI Applications with LangChain (Asimov Academy)                   | Aug 2024 |
| • LangChain Chat with Your Data (DeepLearning.AI)                   | Aug 2024 |
| • ChatGPT Prompt Engineering for Developers (DeepLearning.AI)       | Jul 2024 |
| • Advanced Microsoft Excel (Udemy)                                  | Jan 2022 |
| • Advanced Power BI (Udemy)                                         | Jan 2022 |
| • Statistics with R Language (Udemy)                                | Sep 2021 |
| • III Workshop in Applied Statistics (UNIR)                         | Jul 2020 |